

Problem 1. In a university login system, each student has a profile that stores their name and total number of completed credits. A student's **GPA is calculated by dividing total grade points by completed credits**. However, two issues can occur during the calculation:

1. **Some students may not have created their profiles yet**, so their profile information is missing. When the program tries to access the name or credit information of such students, an unexpected error may occur.
2. **Some students may have completed 0 credits**, but still attempt to calculate GPA. Dividing grade points by zero will generate an invalid mathematical operation that the program must handle safely.

Now, **write** a Java program that uses proper built-in exception classes to prevent the program from crashing for the above two cases and displays a meaningful message for each problematic case and continues processing the remaining students.

Expected Output:

Processing student 1:
Name: Anwar
GPA: 3.75

Processing student 2:
Error: Student profile not found. Skipping this student.

Processing student 3:
Name: Rahim

Problem 2.

Imagine you have a bank account. You can deposit and withdraw money from your account. You should keep in mind that the total amount of money withdrawn from your account must not exceed the total balance present in your account. If such a scenario happens, you need to safely execute from the banking system. **Implement** the above case in Java with the proper utilization of user-defined exception class with throw, throws, try and catch.

Expected Output:

Enter amount to deposit: 10000
Deposit successful! Current balance: 10000.0
Enter amount to withdraw: 2500
Withdrawal successful! Current balance: 7500.0
Enter amount to withdraw: 8000